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Lewis acid-catalyzed divergent synthesis of polysubstituted furans and pyrroles†

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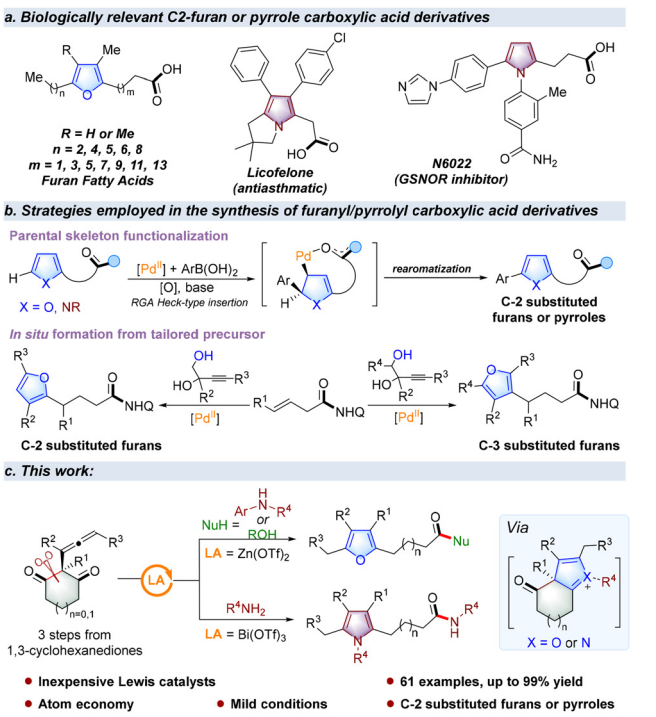
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The development of an advisable strategy to obtain biologically relevant polysubstituted furans and pyrroles is a highly anticipated yet challenging task. Here we report an efficient synthetic strategy for the rapid construction of multisubstituted C2-furan or pyrrole carboxylic acid derivatives *via* a cascade reaction from simple and readily available starting materials. This catalytic protocol, characterized by inexpensive Lewis catalysts and high atom economy, provides access to a wide variety of tri-, and tetrasubstituted furan/pyrrole derivatives in excellent yields. Gram-scale synthesis and synthetic applications are demonstrated.

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Five-membered heterocycles are very important fundamental heterocyclic motifs widely found in diverse applications of organic materials, pharmaceutical science and synthetic chemistry.[1] Among these heterocyclic molecules, densely functionalized furans and pyrroles have garnered a particular interest since they are the primary constituents of many natural products that have special biological activities, pharmaceuticals, and organic materials such as photographic chemicals and optoelectronic materials.[2] In particular, furan derivatives with long-chain carboxylic acids, like furan fatty acids (FuFAs), constitute an important class of biologically relevant natural products due to their anti-inflammatory and cardiovascular activities, affording significant nutritional health benefits to humans (Scheme 1a).[3] *N*-Aryl pyrrole derivatives have also exhibited intriguing bioactivities ranging from antibacterial, antiviral, anti-inflammatory, and antitumoral properties (Scheme 1a).[4] It is consequently not surprising that the development of diverse methods to synthesize functionalized furans and pyrroles is an area of intense research in modern organic chemistry.

The widely reported synthetic methods of these heterocyclic compounds are of great concern and can be broadly classified into two categories.[5] One is the functionalization of an existing furan or pyrrole ring.[6] Another involves the heteroatom-mediated cyclization of an acyclic substrate to construct a heterocyclic ring, with all or most of the desired substituents pre-installed on the precursor, *via* classical methods (Paal–Knorr,[7] Hantzsch,[8] and Feist–Benary[9] reactions) or the present catalytic methods.[10] Thus far, only a few methods to access C2-substituted furanyl carboxylic acid derivatives have been reported through the two strategies mentioned.[11] For example, Kimber and co-workers realized the first total synthesis of the tetrasubstituted furan fatty acid (FFA) metabolite 5-[(1*E*)-2-car-



Scheme 1 Overview.

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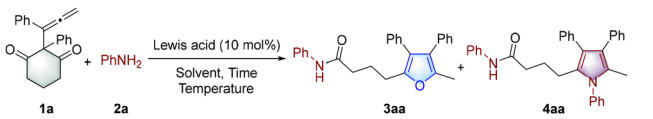
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boxyethenyl]-3,4-dimethyl-2-furanpentanoic acid (CeDFP) *via* Au-catalyzed alkyne hydroarylation in 10 steps (4.1% overall yield).[11e] They also effectively constructed tetrasubstituted furans suitable for furan fatty acid total synthesis with a palladium-catalyzed acid-mediated isomerization sequence in high yield.[11d] Meanwhile, Yin's group reported a mild strategy for oxidative C_{sp^2}–H arylation of electron-rich furans and pyrroles with the aid of remote carbonyl-containing groups (Scheme 1b).[11b] This protocol has been successfully applied to the synthesis of the skeleton of the pyrrole-based GSNOR inhibitor. Volla's group reported a convergent strategy for preparing γ-furan-3-yl carboxylic acid derivatives *via* a Pd(II)-catalyzed intramolecular cascade cyclization of alkyne-diols followed by hydrocarbofunctionalization of unactivated alkenes employing a cleavable 8-aminoquinoline directing group (Scheme 1b).[11a] A complementary protocol for accessing γ-furan-2-yl carboxylic acid derivatives was also developed by carrying out the reaction in a tandem manner.

Although the present methods are efficiently applied to obtain differently substituted furans and pyrroles, these methodologies generally require the use of expensive or air-sensitive transition-metal catalysts and have poor accessibility of the starting materials with limited substrate scopes.[12] Therefore, the development of new strategies that facilitate the rapid construction of furans or pyrroles with multiple functional groups is still a challenging topic, and new protocols for addressing the issue are still highly desirable. As a continuation of our interest in the synthesis of functionalized (benzo)furans through transition metal catalysis,[13] we hereby describe a divergent method involving two inexpensive Lewis acids for the construction of diverse C2-functionalized furanyl/pyrrolyl carboxylic acid derivatives in good yields (Scheme 1c).

We commenced our investigation by exploring the model reaction between β-ketoallene **1a** and aniline **2a**, using 10 mol% Ph_3PAuCl as the catalyst and NaBARF (12 mol%) as a halide scavenger in toluene at room temperature under N_2 for 5 h, which afforded the desired furan product **3aa** in 25% yield and the unexpected pyrrole product **4aa**, albeit in 5% yield (Table 1, entry 1). Significant improvement was not observed with higher temperature, possibly due to the decomposition of substrate **1a** or failure to condense with aniline **2a** (see Table S1† for more details). To our delight, inexpensive Zn $(OTf)_2$ and $Bi(OTf)_3$ showed good selectivity and efficiency to respectively provide **3aa** in 92% yield and **4aa** in 66% yield when we turned to screening different Lewis acid catalysts, although most of them have shown disappointing results (Table 1, entries 2–10). Moreover, decreasing the aniline **2a** load to 1.5 equiv. was also found to be effective in providing **3aa** in 92% yield, while lowering the temperature to 80 °C resulted in a decrease in the yield of **3aa** to 62% (Table 1, entries 11 and 12), and a further decrease in the employed amount of aniline **2a** to 1.2 or 1.0 equiv. caused a slight decrease in the amount of **3aa** (see Table S2 of the ESI†). Although no positive result was obtained by using other solvents like chlorobenzene or benzotrifluoride to improve the efficiency of the protocol for producing **3aa** (see Tables S2 and S3†), we were delighted to observe that cyclohexane rendered the best result for the transformation with **1a** and **2a** employing 10 mol% $Bi(OTf)_3$ to deliver **4aa** in 83% yield (Table 1, entry 13). Interestingly, as the reaction proceeded, we observed that some products were precipitated from the reaction system, revealing one possible reason why cyclohexane is superior to other solvents. The structure of **3aa** was unambiguously confirmed by X-ray diffraction (see the ESI† for details).

Table 1 Optimization of the reaction conditions[a]



1a 2a 3aa 4aa

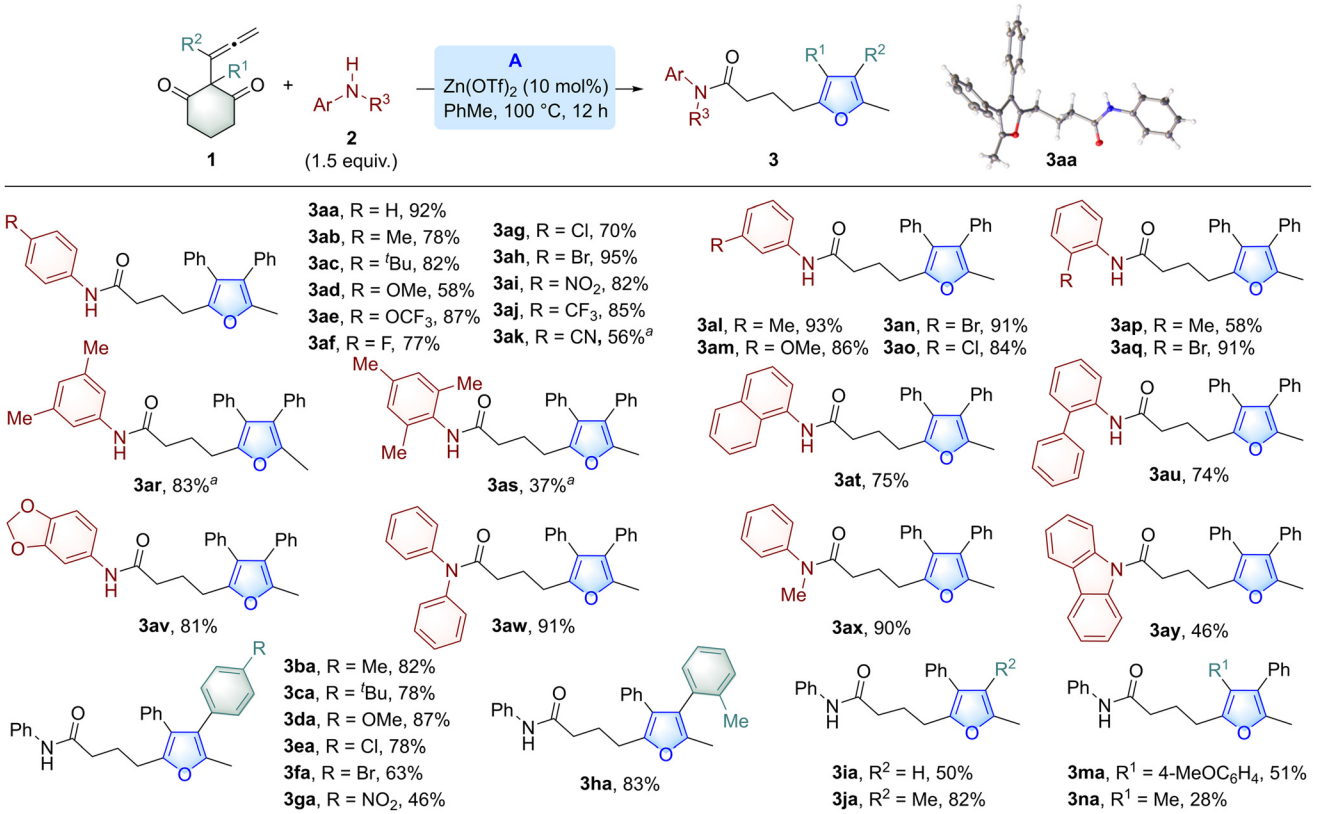
Entry	Lewis acid	Solv./temp. (°C)/time (h)	Conv.[b] (%)	**3aa**[b] (%)	**4aa**[b] (%)
1[c]	PPh_3AuCl	PhMe/RT/12	85	25	5
2	$PtCl_2$	PhMe/100/60	50	13	10
3	$B(C_6F_5)_3$	PhMe/100/60	20	10	5
4	AgOTf	PhMe/100/60	84	50	33
5	$Sc(OTf)_3$	PhMe/100/60	86	7	46
6	$Cu(OTf)_2$	PhMe/100/60	94	29	22
7	$In(OTf)_3$	PhMe/100/60	55	7	42
8	$Yb(OTf)_3$	PhMe/100/60	72	22	5
9	$Bi(OTf)_3$	PhMe/100/60	89	12	70 (66)
10	$Zn(OTf)_2$	PhMe/100/12	100	94 (92)	5
11[d]	**$Zn(OTf)_2$**	**PhMe/100/12**	**100**	**93 (92)**	**1**
12	$Zn(OTf)_2$	PhMe/80/36	83	68 (62)	12
13	**$Bi(OTf)_3$**	**Cyclohexane/100/72**	**99**	**11 (9)**	**85 (83)**

[a] Unless otherwise mentioned, all reactions were performed under N_2 with 10 mol% catalyst, 0.10 mmol of **1a**, 0.30 mmol of **2a**, and PhMe (0.8 mL) at 100 °C. [b] Yields determined by ^1H NMR analysis of the crude reaction mixtures using 4-nitrotoluene as an internal standard. Isolated yield in parentheses. [c] NaBARF (10 mol%) was added. [d] **2a** (1.5 equiv.).

Further experiments were then conducted to explore the generality of the cascade reaction concerning a variety of aromatic amines with cyclic β-ketoallenes. As shown in Scheme 2, a variety of multisubstituted furans were competently synthesized. The protocol first succeeded with a variety of primary anilines bearing *para* substituents, whereas the electronic perturbation of substituents had a distinct influence on the reaction efficiency. Aniline, 4-methyl aniline, and 4-*tert*-butyl aniline engaged in reactivity using β-ketoallene **2a**, with the desired furans **3aa–3ac** formed in 78–92% isolated yields. For the *para*-methoxy variant, the desired product **3ad** was furnished in 58% yield. The conversion of 4-halogeno anilines with compound **2a** also proceeded smoothly, delivering products **3af–3ah** in good isolated yields (70–95%). Gratifyingly, the anilines having different electron-withdrawing groups at the *para* position, such as trifluoromethoxy, nitro and trifluoromethyl, were also efficiently converted into the corresponding products **3ae–3aj** in 87%, 82% and 85% yields, respectively, while 4-cyano aniline slowly transformed into **3ak** in 56% yield with 30 h of heating. Meanwhile, the *ortho*-methyl and -phenyl substituted anilines could also be tolerated, generating the corresponding products **3ap** and **3au** in slightly lower yields of 58%

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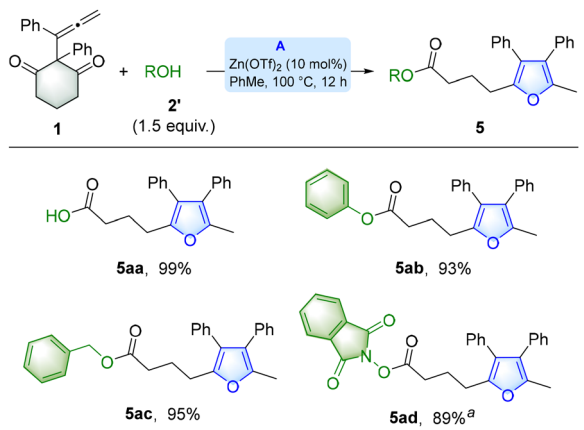


Scheme 2 Substrate scope of polysubstituted furans. Reaction conditions: unless otherwise noted, all reactions were carried out with **1** (0.2 mmol, 1.0 equiv.), **2** (0.3 mmol, 1.5 equiv.), and $Zn(OTf)_2$ (10 mol%) in PhMe (1.5 mL) at 100 °C for 12 h. Products **3** were obtained in isolated yields. [a] 30 h.

and 74%. However, other substituents at either the *ortho* or *meta* position could also afford the corresponding products **3al–3aq** in excellent yields (84–93%). In addition, di- and multi-substituted anilines **2r** and **2s** can also obtain the target products with distinct yields (83% and 37%) due to the significant steric hindrance. 1-Naphthylamine **2t** bearing a strong conjugated aromatic system and 3,4-(methylenedioxy)aniline **2v** could also offer the target products in good yields (**3at** and **3av**, 75–81%). Particularly, changing primary aromatic amines to secondary aromatic amines was compatible with the synthesis of the desired products **3aw–3ay** in moderate to good yields (46–91%).

Subsequently, we turned our attention to defining the capacity of different cyclic β-allenyl 1,3-diketones in this reaction to react with aniline. With respect to the substituents at the R^2 position, both electron-rich and electron-deficient substituents on the phenyl group (**1b–1h**) could be employed for this transformation, and the corresponding tetrasubstituted furans **3ba–3ga** were achieved in modest to good yields (46–87%). The nature of the substituents had a demonstrable effect on this reaction. As for the electron-withdrawing 4-nitro and 4-Br groups, **3ga** and **3fa** were afforded in modest yields. Moreover, product **3ha** containing the sterically hindered 2-methylphenyl substituent was provided in 83% yield. It is noted that when monosubstituted allene **1i** was tested, the reaction provided the corresponding furan product **3ia** in 50% yield. When R^2 was methyl in the allene substrate, **3ja** could still be obtained in 82% yield.

Next, a series of oxygen nucleophiles was then explored under identical conditions to synthesize various tetrasubstituted furan derivatives (Scheme 3). C2-substituted furan car-



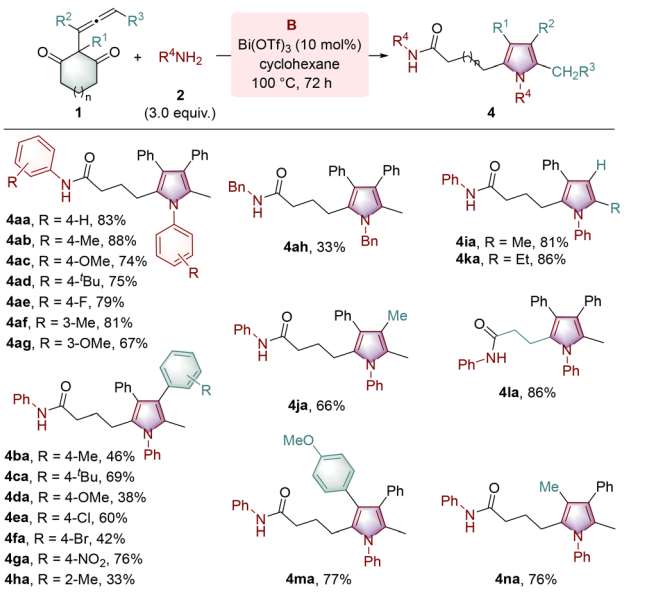
Scheme 3 Scope of oxygen nucleophiles. Reaction conditions: unless otherwise noted, all reactions were carried out with **1** (0.2 mmol, 1.0 equiv.), **2** (0.3 mmol, 1.5 equiv.), and $Zn(OTf)_2$ (10 mol%) in PhMe (1.5 mL) at 100 °C for 12 h. Products **5** were obtained in isolated yields. [a] 30 h.

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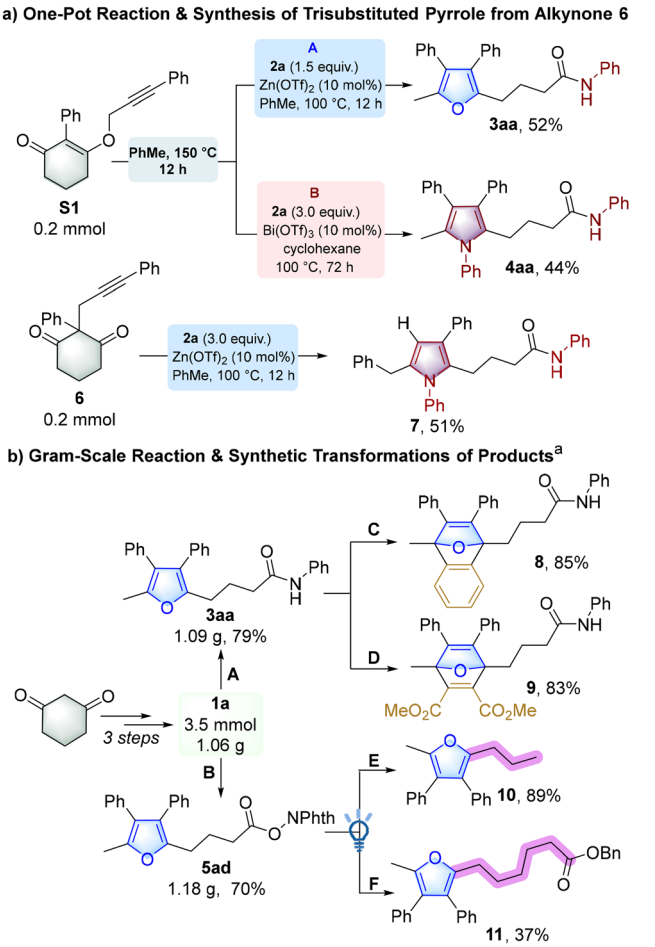
boxylic acid **5aa** was obtained quantitatively in the presence of H_2O. At the same time, three different ester derivatives **5ab**–**5ad** could be separated in high yields (89–95%) with phenol, benzyl alcohol and *N*-hydroxyphthalimide as nucleophiles.

Similarly, with the corresponding optimal reaction conditions in hand, we next explored the substrate scope for the synthesis of pyrroles (Scheme 4). Overall, various substituted aromatic amine compounds with electron-donating and -withdrawing groups could undergo reactions smoothly with moderate to good yields (67–88%). To our delight, benzylamine **2h** could also be tolerated with generating product **4ah** in 33% isolated yield. The effect of different substituents on the allene substrates was investigated. It was found that various substituents on the phenyl group of R^2 had an impact on the reactions. Electron-donating groups resulted in relatively lower yields than electron-withdrawing ones (**4ba**–**4ga**, 38–76%). It is noted that when the methyl group was placed at the *ortho* position of the phenyl group, the reaction gave the desired product **4ia** in 33% yield. When monosubstituted allene **1i** and 1,3-disubstituted allene **1k** were utilized as the substrates, the reaction gave products **4ia** and **4ka** in 81% and 86% yields, respectively. When R^2 was a methyl group and R^3 was H, the reaction still proceeded smoothly to afford the desired product in good yield (**4ja**, 66%). β-Ketoallene **1l** with a 1,3-cyclopentanedione ring was investigated as well, successfully providing the desired product **4la** in 86% yield.

We were glad to find that the use of cyclohexenone derived **S1** could realize a one-pot synthesis of tetrasubstituted furan derivative **3aa** in 52% yield and pyrrole derivative **4aa** in 44% yield (Scheme 5a). Furthermore, a distinctive trisubstituted



Scheme 4 Substrate scope of polysubstituted pyrroles. Reaction conditions: unless otherwise noted, all reactions were carried out with **1** (0.2 mmol, 1.0 equiv.), **2** (0.6 mmol, 3.0 equiv.), and $Bi(OTf)_3$ (10 mol%) in cyclohexane (1.8 mL) at 100 °C for 72 h. Products **4** were obtained in isolated yields.

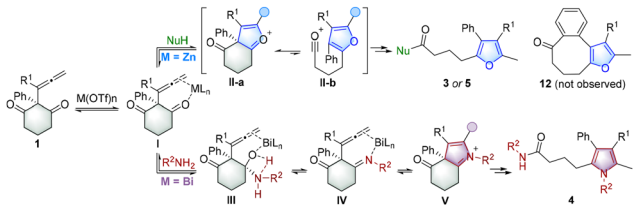


Scheme 5 Further investigations. [a] Condition **A**: **1a** (3.5 mmol, 1.0 equiv.), **2a** (1.5 equiv.), and $Zn(OTf)_2$ (10 mol%) in PhMe (20 mL) at 100 °C for 36 h; condition **B**: **1a** (3.5 mmol, 1.0 equiv.), *N*-hydroxyphthalimide (1.5 equiv.), and $Zn(OTf)_2$ (10 mol%) in PhMe (20 mL) at 100 °C for 36 h; condition **C**: aryne precursor, CsF, CH_3CN, 80 °C, 4 h; condition **D**: DMAD, 1,4-dioxane, 110 °C, 6 h; condition **E**: HE, DMA, blue LEDs (456 nm), r.t., 24 h; condition **F**: benzyl acrylate, HE, DMA, blue LEDs (456 nm), r.t., 24 h.

furan **7** was also obtained from the reaction of alkynone **6** and aniline under the catalysis of $Zn(OTf)_2$ in 51% yield (Scheme 5a). To further explore the efficiency, practicality, and generality of the reaction protocol presented here, relevant studies were conducted (Scheme 5b). First, the furan products **3aa** and **5ad** could also be obtained in comparable yields (79% and 70%) on a gram scale (3.5 mmol). Downstream transformations of the furan moiety were then carried out. Through a [4 + 2] cycloaddition,[13g] treatments of **3aa** with dienophiles, such as DMAD and benzyne, produced Diels–Alder adducts **8** and **9** in good yields. Decarboxylative conjugate addition and hydrodecarboxylation of aliphatic *N*-(acyloxy)phthalimide **5ad** were also performed through photoirradiation with Hantzsch ester in *N*,*N*-dimethylacetamide solution,[14] furnishing the corresponding products **10** and **11** in 89 and 37% yields, respectively.

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Scheme 6 Proposed mechanism.

A plausible reaction mechanism for this cascade protocol has been proposed in Scheme 6. Initially, β-ketoallene **1** is activated by Lewis acid catalysts to form a dual-coordinated adduct **I** with the allene moiety and the oxygen of the carbonyl group, which display distinct activity depending on the catalytic metals. The catalytic zinc center might slightly favour coordination to the allene moiety over the oxygen, thus triggering an intramolecular 5-*exo-trig* cyclization to form an oxonium intermediate **II-a**. In the absence of nucleophiles, the reaction is almost shut down with a large amount of β-ketoallene **1** being preserved, and no Friedel–Crafts cyclization product **12** is observed. Therefore, we postulated that the resulting **II-a** is sluggish to undergo fragmentation to form the acyl cation **II-b**. If a very small amount of transient **II-b** is formed, it would be trapped by excessive external nucleophiles in an irreversible manner, thus driving the processes toward the desired product. In contrast, the bismuth catalyst might have a stronger affinity to oxygen than the allene moiety, resulting in a more active carbonyl group, which is subsequently captured by aniline to give geminal amino alcohol **III** and then undergoes bismuth-assisted dehydration to form ketimine **IV**.[12a,15] Finally, the more nucleophilic ketimine engages the activated allene moiety through a similar 5-*exo-trig* cyclization, followed by a fragmentation–trapping process to furnish pyrrole **4**.

Conclusions

In summary, we have developed an operationally simple, unified strategy for the synthesis of diversely substituted C2-furan or pyrrole carboxylic acid derivatives *via* a cascade reaction from simple and readily available/preparable starting materials. The cascade is proposed to go through consecutive processes of cyclization, ring fragmentation, aromatization and an S_N1 reaction involving acyl cation species as the key intermediate. This catalytic protocol, characterized by inexpensive catalysts and high atom economy, provides access to a wide variety of tri- and tetrasubstituted furan/pyrrole derivatives in excellent yields. In addition, the cyclization–fragmentation strategy was studied for its generality and scalability by carrying out gram-scale synthesis and downstream synthetic transformations. This methodology might also provide a potential synthetic route to tetrasubstituted furan fatty acids.

Data availability

The data supporting this article have been included as part of the ESI.†

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

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